Project: Worked examples
Title: Helical springs: compression, extension, torsion
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1. Scope and method

A cylindrical helical spring of round wire, cold-coiled, with a linear characteristic: compression (rate, working points, solid length by end type, stress at the working points and at solid, buckling, Goodman fatigue, surge frequency), extension (the same with the initial tension and the hook stresses) or torsion (rate with the legs, curvature-corrected bending stress, wound diameter and body length).
Index and rate: C = D/d; k = G·d⁴/(8·D³·n) (compression, extension); k_φ = E·d⁴/(64·D·n_a), n_a = n + (l₁ + l₂)/(3π·D) (torsion) Stress: τ_k = k_c·8·F·D/(π·d³), k_c = (C + 0.5)/(C − 0.75) (Bergsträsser) or (4C − 1)/(4C − 4) + 0.615/C (Wahl); torsion σ = K_i·32·M/(π·d³), K_i = (4C² − C − 1)/(4C·(C − 1)) Solid length (compression, derived from the geometry of the end): n_t = n + e_n, L_s = d·(n_t + e_s) with (e_n, e_s) = (0, 1) plain, (1, 0) plain and ground, (2, 1) squared, (2, 0) squared and ground; pitch p = (L₀ − a·d)/(n + b) Extension: F = F₀ + k·s; hook bending σ_A = F·[K_A·16·D/(π·d³) + 4/(π·d²)], K_A = (4C₁² − C₁ − 1)/(4C₁·(C₁ − 1)), C₁ = 2r₁/d; hook torsion τ_B = K_B·8·F·D/(π·d³), K_B = (4C₂ − 1)/(4C₂ − 4), C₂ = 2r₂/d Buckling (compression): λ_eff = ν·L₀/D; y_cr = L₀·C₁′·[1 − √(1 − C₂′/λ_eff²)], C₁′ = E/(2(E − G)), C₂′ = 2π²(E − G)/(2G + E); stable at any deflection when λ_eff² < C₂′ Fatigue (Goodman, as M4): 1/n_f = a/S_e + m/S_u with a = (x₂ − x₁)/2, m = (x₂ + x₁)/2 of the stresses at the two working points; S_u = f_su·R_m in shear, R_m in bending Surge: f_n = (d/(2π·n·D²))·√(G/(2ρ)); warning when f_n/f_op < 13 Torsion body: D_w = D·n/(n + φ/2π); L_w = d·(n + 1 + φ/2π)

Ref: EN 13906-1 Calculation of compression springs — Rate, stress correction, solid length, buckling, fatigue, surge
Ref: EN 13906-2 Calculation of extension springs — Rate, initial tension, hook stress
Ref: EN 13906-3 Calculation of torsion springs — Rate, bending stress with curvature correction
Ref: Shigley's Mechanical Engineering Design 10th ed., Ch. 10 — Helical springs: end geometry, buckling, critical frequency, extension and torsion springs

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Kind of spring	Compression		spec: Plunger assembly PLG-210 §4 rev B	any
corr	Stress correction factor k_c for the shear stress	Bergsträsser		spec: Company spring standard SPR-01 (EN 13906-1 correction factor)	any
d	Wire diameter	3	mm	drawing: PLG-210-04 rev B	0.1 … 20 mm
D	Mean coil diameter	24	mm	drawing: PLG-210-04 rev B	0.5 … 400 mm
n	Active coils (torsion: coils of the body)	9		drawing: PLG-210-04 rev B	2 … 200
ends	End type (compression)	Squared and ground		drawing: PLG-210-04 rev B	any
L0	Free length (compression)	70	mm	drawing: PLG-210-04 rev B	1 … 1000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
work	Working points given as	Loads		spec: Plunger assembly PLG-210 §4 rev B	any
F_1	Force at working point 1 (loads)	60	N	spec: Plunger assembly PLG-210 §4 rev B	0 … 20000 N
F_2	Force at working point 2 (loads)	180	N	spec: Plunger assembly PLG-210 §4 rev B	0 … 20000 N
G	Shear modulus	79.3	GPa	assumption: Generic spring steel shear modulus; take it from the W3 material of the wire certificate.	20 … 100 GPa
E	Young's modulus	206	GPa	assumption: Generic spring steel Young's modulus; take it from the W3 material of the wire certificate.	50 … 250 GPa
rho	Density	7850	kg/m^3	assumption: Generic steel density.	1000 … 20000 kg/m^3
Rm	Tensile strength of the wire at this diameter R_m(d)	1650	MPa	assumption: Example value for a 3 mm wire. Enter R_m(d) from your own copy of the wire standard or the wire certificate (record spring.wire-strength); the standard's table is not shipped.	300 … 3500 MPa
f_tau	Permissible shear stress as a fraction of R_m	0.45		assumption: Static permissible shear stress fraction of R_m for the example; take it from your design reference (record spring.permissible).	0.2 … 0.8
f_su	Ultimate shear strength as a fraction of R_m (Goodman, shear)	0.67		assumption: Ultimate shear strength fraction of R_m used by the Goodman relation (textbook steel value); confirm in your reference.	0.4 … 0.9

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
nu_seat	End-support coefficient v (0.5 both ends fixed, 0.7 one fixed one pinned, 1 both pinned, 2 one free)	0.5		spec: Plunger assembly PLG-210 §4: both end plates flat and parallel, no rotation	0.5 … 2
S_e	Endurance strength for the stress kind (shear; bending in torsion) at N_c cycles, alternating	450	MPa	assumption: Example alternating shear endurance strength at 10^7 cycles; take it from the wire maker or your fatigue reference (record spring.endurance).	50 … 2000 MPa
N_c	Required life the endurance strength belongs to	10000000		spec: Plunger assembly PLG-210 §4: duty life	10000 … 10000000000
n_f_req	Required Goodman safety factor	1.2		spec: Company spring standard SPR-01	1 … 5
f_op	Operating (excitation) frequency	8	Hz	spec: Plunger assembly PLG-210 §4: maximum cycle rate	0 … 1000 Hz

3. Assumptions

A1. Linear characteristic: the rate is constant between the working points; no progressive or conical spring, no variable pitch. (Cold-coiled round wire up to 20 mm.)
A2. Active coils n are as entered; end coils are accounted for only in the solid length (and the pitch) by the end type; the hook and leg flexibility is in the rate only for the torsion legs. (Shigley 10th ed. Ch. 10, geometry of the ends.)
A3. R_m(d), the permissible stress fractions and the endurance strength are the user's inputs from the user's own copy of the wire standard and design reference; no wire-strength table and no permissible-stress diagram is shipped. (Addendum B §B5.)
A4. The stress correction factor k_c is applied to the working and the solid stress alike (conservative for the static check). (Choose 'None (uncorrected)' for the uncorrected static stress.)
A5. Extension: hook fatigue and the hook shape beyond its two radii are not assessed; the initial tension is a force the user enters. (Shigley 10th ed. Ch. 10, extension springs.)
A6. Surge is the first natural frequency of a spring with both ends fixed; a spring with one end free has half of it. (Shigley 10th ed. Ch. 10, critical frequency.)
A7. G (Shear modulus) is taken as 79.3 GPa. (Generic spring steel shear modulus; take it from the W3 material of the wire certificate.)
A8. E (Young's modulus) is taken as 206 GPa. (Generic spring steel Young's modulus; take it from the W3 material of the wire certificate.)
A9. rho (Density) is taken as 7850 kg/m^3. (Generic steel density.)
A10. Rm (Tensile strength of the wire at this diameter R_m(d)) is taken as 1650 MPa. (Example value for a 3 mm wire. Enter R_m(d) from your own copy of the wire standard or the wire certificate (record spring.wire-strength); the standard's table is not shipped.)
A11. f_tau (Permissible shear stress as a fraction of R_m) is taken as 0.45. (Static permissible shear stress

fraction of R_m for the example; take it from your design reference (record spring.permissible).)
A12. f_su (Ultimate shear strength as a fraction of R_m (Goodman, shear)) is taken as 0.67. (Ultimate shear strength fraction of R_m used by the Goodman relation (textbook steel value); confirm in your reference.)
A13. S_e (Endurance strength for the stress kind (shear; bending in torsion) at N_c cycles, alternating) is taken as 450 MPa. (Example alternating shear endurance strength at 10^7 cycles; take it from the wire maker or your fatigue reference (record spring.endurance).)

4. Working

[6] C = D / d — EN 13906-1 / EN 13906-2 / EN 13906-3
= 24 mm / 3 mm
= 8
Spring index

[14] k_c = (C + 0.5) / (C - 0.75)
= (8 + 0.5) / (8 − 0.75)
= 1.1724
Shear stress correction factor (Bergsträsser)

[15] tau_perm = f_tau * Rm
= 0.45 × 1650 MPa
= 742.5 MPa
Permissible shear stress (fraction of R_m, user input)

[17] k = G * d^4 / (8 * D^3 * n) — EN 13906-1
= 79.3 GPa × (3 mm)^4 / (8 × (24 mm)^3 × 9)
= 6.4535 N/mm
Spring rate

[18] n_t = n + 2
= 9 + 2
= 11
Total coils (derived: end coils added to the active coils)

[19] L_s = d * (n_t + 0) — EN 13906-1
= 3 mm × (11 + 0)
= 33 mm
Solid length (derived from the end type)

[20] p = (L0 - 2 * d) / (n + 0)
= (70 mm − 2 × 3 mm) / (9 + 0)
= 7.1111 mm
Pitch of the free spring (derived)

[21] s_c = L0 - L_s
= 70 mm − 33 mm
= 37 mm
Travel to solid

[22] F_c = k * s_c
= 6.4535 N/mm × 37 mm
= 238.78 N
Force at solid length

[26] s_1 = F_1 / k
= 60 N / 6.4535 N/mm
= 9.2974 mm
Deflection at working point 1

[27] s_2 = F_2 / k
= 180 N / 6.4535 N/mm
= 27.892 mm
Deflection at working point 2

[28] tau_1 = k_c * 8 * F_1 * D / (pi * d^3) EN 13906-1
= 1.1724 × 8 × 60 N × 24 mm / (π × (3 mm)^3)
= 159.23 MPa
Shear stress at working point 1
[29] tau_2 = k_c * 8 * F_2 * D / (pi * d^3) EN 13906-1
= 1.1724 × 8 × 180 N × 24 mm / (π × (3 mm)^3)
= 477.68 MPa
Shear stress at working point 2
[30] tau_c = k_c * 8 * F_c * D / (pi * d^3) EN 13906-1
= 1.1724 × 8 × 238.78 N × 24 mm / (π × (3 mm)^3)
= 633.67 MPa
Shear stress at solid length
[31] u_2 = tau_2 / tau_perm
= 477.68 MPa / 742.5 MPa
= 0.64335
Utilisation at working point 2
[32] u_c = tau_c / tau_perm
= 633.67 MPa / 742.5 MPa
= 0.85343
Utilisation at solid length
[33] L_1 = L0 - s_1
= 70 mm − 9.2974 mm
= 60.703 mm
Spring length at working point 1
[34] L_2 = L0 - s_2
= 70 mm − 27.892 mm
= 42.108 mm
Spring length at working point 2
[35] lambda_eff = α·L_b/D EN 13906-1
= 0.5 × 70 / 24
= 1.4583
Effective slenderness
[36] C1p = E/(2(E - G)) Shigley 10th ed. Ch. 10, buckling
= 206 / (2 × (206 − 79.3))
= 0.81294
Buckling constant C_1'
[37] C2p = 2π²(E - G)/(2G + E) Shigley 10th ed. Ch. 10, buckling
= 2π² × (206 − 79.3) / (2 × 79.3 + 206)
= 6.8595
Buckling constant C_2'
[38] u_b = stable at any deflection (α_eff² < C₂')
= 2.1267 < 6.8595
= 0
Deflection at point 2 against the buckling deflection (none: stable)
[39] f_n = d / (2 * pi * n * D^2) * sqrt(G / (2 * rho)) EN 13906-1 / EN 13906-2
= 3 mm / (2 × π × 9 × (24 mm)^2) × sqrt(79.3 GPa / (2 × 7850 kg/m³))
= 207 Hz
Surge (natural) frequency, both ends fixed
[41] r_f = f_n / f_op
= 207 Hz / 8 Hz
= 25.875
Surge frequency / operating frequency

[46] tau_a = (tau_2 - tau_1) / 2
= (477.68 MPa - 159.23 MPa) / 2
= 159.23 MPa
Alternating shear stress
[47] tau_m = (tau_2 + tau_1) / 2
= (477.68 MPa + 159.23 MPa) / 2
= 318.46 MPa
Mean shear stress
[48] S_su = f_su * Rm
= 0.67 × 1650 MPa
= 1105.5 MPa
Ultimate shear strength (fraction of R_m, user input)
[49] u_f = tau_a/S_e + tau_m/S_su
= 159.23 / 450 + 318.46 / 1105.5
= 0.64191
Goodman, as M4 (Shigley 10th ed. Ch. 6)
Goodman utilisation (1/n_f)
[50] n_f = 1/u_f
= 1 / 0.64191
= 1.5579
Goodman safety factor
[51] u_fat = n_f_req * u_f
= 1.2 × 0.64191
= 0.77029
Fatigue utilisation against the required factor

5. Results

SYMBOL	VALUE	DESCRIPTION
C	8	Spring index
k	6.4535 N/mm	Rate
k_c	1.1724	Stress correction factor
n_t	11	Total coils
L_s	33 mm	Solid length
p	7.1111 mm	Pitch
F_1	60 N	Force at point 1
F_2	180 N	Force at point 2
s_1	9.2974 mm	Deflection at point 1
s_2	27.892 mm	Deflection at point 2
L_1	60.703 mm	Length at point 1
L_2	42.108 mm	Length at point 2
s_c	37 mm	Travel to solid
F_c	238.78 N	Force at solid
tau_1	159.23 MPa	Stress at point 1
tau_2	477.68 MPa	Stress at point 2
tau_c	633.67 MPa	Stress at solid
u_2	0.64335	Utilisation at point 2

SYMBOL	VALUE	DESCRIPTION
u_c	0.85343	Utilisation at solid
y_cr	"— (stable at any deflection)"	Buckling deflection
u_b	0	Deflection at point 2 / buckling deflection
buckling	"PASS"	Buckling
f_n	207 Hz	Surge frequency
r_f	25.875	Surge ratio f_n/f_op
u_f	0.64191	Goodman utilisation
n_f	1.5579	Goodman safety factor

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Static shear stress at solid length	u_c = 0.85343	u_c ≤ 1	0.853	EN 13906-1
PASS	Goodman fatigue against the required safety factor	u_fat = 0.77029	u_fat ≤ 1	0.77	Goodman, as M4
PASS	Working deflection within the travel to solid	s_2 = 27.892 mm	s_c = 37 mm	0	EN 13906-1

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: G, E, rho (Material properties); Rm (Wire tensile strength R_m(d)); f_tau, f_su (Permissible and ultimate stress fractions); nu_seat (End-support coefficient); S_e (Endurance strength). Enter each as one so its source, edition and licence travel with it.

8. Verification

Template mech.spring@1.0.0 Content hash 45fc6646c6a23b96 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0